

Measuring Rhetorical Alarmism in Disaster News: A Quantitative Framework Applied to the 2024 Valencia Floods

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Abstract

1 Introduction

The floods that affected the Valencia region in October 2024 have been described as one of the most severe natural disasters in recent Spanish history, resulting in 228 fatalities and affecting thousands of people. In such contexts, news media play a crucial role in society, as they are responsible for disseminating information and shaping how events are publicly interpreted (Pantti et al. 2012; Cottle 2014). For this reason, media coverage contributes to the construction of meaning around disasters, influencing not only how they are understood but also how they are socially and politically responded to.

Disasters are not only objective events defined by their material consequences, but also socially constructed phenomena influenced by the media representations (Pantti et al. 2012). The way in which different media outlets select, frame and narrate these events, plays a critical role in defining their salience, perceived severity, and their social implications. Rather than acting as neutral transmitters of information, media operate as *'amplification stations'* of risk, intensifying certain aspects of the event while downplaying others, thereby influencing public perception and collective responses (Kasperson et al. 1988). This can be further understood through the framing theory, which posits that media do not simply report events but actively construct narratives by selecting, emphasizing and organizing specific aspects of reality (Scheufele 1999). Frames operate by defining problems, attributing causes and suggesting moral evaluations or potential solutions (Entman 1993).

However, the process of risk amplification is not neutral and may lead to distortions in how crises are represented and understood. In highly mediatized disaster contexts, the selection and intensification of specific elements can give rise to narratives that emphasize threat, urgency, and uncertainty in disproportionate ways (Cottle 2012). Holm (2012) listed an excellent example of this phenomenon regarding the media coverage of Katrina Hurricane, the *New York Times* wrote: *"America is once more plunged into a snakepit of anarchy, death, looting, raping, marauding thugs, suffering infrastructure, a gutted police force, insufficient troop levels, and criminally negligent government planning"*. This is particularly evident in alarmist headlines, where linguistic and rhetorical choices tend to condense and dramatize complex events, often privileging emotionally charged interpretations over contextual nuance. As noted by Molek-Kozakowska (2013), headlines work as strategic devices that package information, activate emotional and cognitive responses, and frame the audience's understanding of the event. As a result, media coverage may contribute to the construction of alarmist narratives that frame

disasters as unprecedented, uncontrollable, or escalating phenomena, potentially amplifying public fear and shaping collective responses beyond the immediate material impact of the event (Altheide and Michalowski 1999; Altheide 2018).

Despite extensive research on media framing and risk communication in the context of natural disasters, existing studies tend to approach *alarmism* in broad or implicit terms, without providing a systematic framework to identify and quantify it (see next section). In particular, while previous work has examined emotional framing, fear appeals, sensationalism, and media hype in crisis contexts, relatively limited attention has been paid to the specific rhetorical patterns through which alarmist narratives are constructed. Importantly, not all negatively framed or emotionally charged headlines can be considered alarmist, highlighting the need for a more precise conceptualization of this phenomenon.

This article introduces the concept of *rhetorical alarmism* as a multidimensional form of media framing that operates through identifiable linguistic and narrative structures. Building on existing literature on framing and risk amplification, the study proposes a framework to systematically capture how media discourse constructs disasters as exceptional, uncontrollable, or escalating events.

Focusing on the case of the 2024 Valencia floods (DANA), the article aims to identify and quantify the presence of *rhetorical alarmism* in news headlines across different Spanish media outlets and social media platforms using an automated algorithm and an ad-hoc index, identifying statistical differences between media types, and examine the temporal evolution of the index.

2 Literature Review

Media play a central role in shaping public understanding of disasters, as well as in structuring broader aspects of social life. As argued by Thompson (1995), communication media transform the spatial and temporal organization of social interaction, reconfiguring relations of visibility and power. In the context of disasters, this transformative role is particularly salient, as media do not simply transmit information about disruptive events but actively participate in their social construction.

In contemporary global communication environments, disasters can no longer be understood as isolated or purely natural phenomena. Instead, they are embedded within transnational communication flows that amplify their visibility and connect distant audiences. In this sense, disasters are increasingly *mediatized*, as media practices shape both the perception and the meaning of such events. Media effectively “enter into” disasters, shaping them from the *inside out* and the *outside in* by reconfiguring social relations and infusing events with cultural and political significance (Pantti et al. 2012; Cottle 2012).

This process is closely linked to the nature of news as discourse. Following Van Dijk (1988), news should be understood not as a neutral representation of reality but as a form of public discourse that structures, interprets, and hierarchizes information. Through its linguistic, semantic, and rhetorical organization, news discourse selectively constructs meanings, embeds ideological perspectives, and guides the interpretation of events. In this sense, news does not merely reflect disasters but actively contributes to the construction of a shared *social imaginary*.

Within this framework, framing emerges as a mechanism through which this construction operates. Framing refers to the process of selecting certain aspects of a perceived reality and making them more salient in communicative texts, in such a way that they promote a particular definition of the problem, a causal interpretation, a moral evaluation, and/or a treatment recommendation (Entman 1993). In the context of disasters, framing has significant implications for audience perception. As noted by Chong and Druckman (2007), the frames adopted in media coverage can lead individuals to develop distinct interpretations of the same issue, influencing the relative weight assigned to different considerations in the formation of opinions.

While framing plays a decisive role in shaping how disasters are interpreted, media coverage also incorporates a strong emotional dimension. As argued by Altheide (1997), the “problem frame” promotes a discourse of fear, understood as the pervasive communication, symbolic awareness, and expectation that danger and risk are central features of the social environment. In this sense, media

discourse not only organizes the interpretation of events but also contributes to the affective construction of public problems. Previous research has shown that public understandings of social issues often incorporate the definitions, scenarios, and symbolic vocabularies circulated by the media, which become part of broader interpretative frameworks through which audiences make sense of risk and crisis (Hall 1980; Vreese 2005; Nisbet 2009).

This is particularly relevant in disaster contexts, where emotionalization can affect how risk, urgency, and severity are interpreted by the public. Moreover, media do not simply reflect risk but can amplify its social perception. According to Kasperson et al. (1988), this amplification may occur through at least three main mechanisms: first, an increased volume of coverage may heighten the salience of a risk event, leading audiences to perceive it as more important, severe, or urgent; second, the presence of expert disagreement or public controversy may intensify uncertainty and concern; and third, risk signals may be amplified when they are communicated in emotionally intense or symbolically charged terms, such as catastrophic metaphors, alarmist language, or highly dramatic imagery. Together, these processes help explain how disaster coverage can move beyond information and toward intensified constructions of threat.

Headlines constitute the primary entry point to a news story, functioning as the first discursive device aimed at capturing the reader’s attention (Kuiken et al. 2017). For this reason, language use at the headline level is not trivial. A headline must condense a substantial amount of information into very few words while simultaneously providing an initial interpretive cue that encourages the reader to engage with the reported event acting as *relevance optimizers* (Dor 2003). In the context of digital journalism and social media, this function becomes even more consequential. The information overload that characterizes contemporary media environments means that headlines, together with visual resources, operate as attention-grabbing mechanisms. Yet it is language that ultimately performs this function, by selecting, intensifying, or dramatizing particular aspects of the event. In this regard, Molek-Kozakowska (2013) proposes a pragma-linguistic framework for the study of sensationalism in news headlines, identifying resources such as specific illocutions, semantic macrostructures, narrative formulas, and evaluative parameters that enhance the perceived relevance, extraordinariness, and intensity of news items. Our study follows a related but more specific concept. Rather than examining sensationalism in general terms, we seek to identify concrete linguistic structures that make it possible to recognize *alarmist* headlines.

Although the literature has extensively examined disaster coverage through the lenses of framing, emotionalization, risk amplification, and sensationalism, alarmism remains a relatively fragmented concept. Existing studies often approach it implicitly, as a by-product of fear, negativity, or media hype, rather than as a distinct discursive configuration. Not all sensationalist or emotionally charged headlines can be considered alarmist. A headline may be dramatic, negative, or affectively loaded without necessarily constructing the event as an exceptional, uncontrollable, or escalating threat. Consequently, it remains unclear which specific linguistic and narrative structures make a headline alarmist, as opposed to merely emotional or sensational. To address this gap, we propose a definition of *rhetorical alarmism* as a multidimensional form of media discourse.

3 A framework for *rhetorical alarmism*

As discussed in the previous section, media do not only summarize events in a neutral way through their headlines. Rather, headlines frame events, activate emotional responses, and amplify the perceived relevance of specific occurrences. In the context of natural disasters or other high-impact social events, these mechanisms become especially relevant, as perceptions of risk, urgency, and threat may be significantly intensified. By using this background, this section introduces our operational definition of *rhetorical alarmism*, grounded in a set of characteristic linguistic resources associated with historical exceptionality, systemic collapse, loss of control or inevitability, and future escalation.

Before introducing the four dimensions of our definition, it is important to highlight that not all emotionally charged, negative, or sensational headlines are necessarily alarmist. A headline may

report a highly severe event and still remain descriptive rather than alarmist. For instance, “*The Valencia floods have left hundreds dead and affected thousands of people*” communicates factual gravity, but it does not frame the event as historically exceptional, systemically collapsing, beyond control, or explicitly escalating into a worse scenario. Rhetorical alarmism therefore is not defined by severity alone, but in a specific rhetorical configuration that amplifies the event beyond its factual description. Under our operational definition, rhetorical alarmism is understood as a form of discourse that disproportionately amplifies the risk, gravity, management or escalation potential of an event through the language of news headlines. In this sense, rhetorical alarmism does not simply describe an event as severe, but constructs it as an exceptional and intensified threat. Considering our previous example, it can be turned into an alarmist headline not by changing the factual core of the event, but by introducing rhetorical ‘elements’ that emphasize exceptionality, a systemic or total collapse, loss of control, or future deterioration. Following our example, a rhetorical-alarmist version of the headline would be: “*The Valencia floods have left hundreds dead and thousands affected in one of the worst disasters in recent Spanish history*” By describing the event as a historical exception, this version goes beyond factual reporting and introduces a rhetorical amplification that frames the event as something extraordinary rather than merely severe. From this perspective, the definition of rhetorical alarmism can be operationalized through four linguistic dimensions that capture its most characteristic forms.

Historical Exceptionality refers to headlines that frame an event as unprecedented, historically unmatched, or extraordinary in relation to prior experience. This dimension is grounded in framing theory, which explains how media select and emphasize particular aspects of reality, and in the literature on sensational headlines, where extraordinariness is often used to heighten salience and reader attention (Entman 1993; Kasperson et al. 1988; Molek-Kozakowska 2013). Consequently, the event is not only presented as severe, but as exceptional in historical terms. A headline such as “The Valencia floods become one of the worst disasters in recent Spanish history” illustrates this dimension because it does not merely report damage; it places the event in a superlative historical frame that amplifies its symbolic weight.

Secondly, **Systemic Collapse** captures headlines that represent the disaster as a breakdown of institutions, infrastructure, or basic social functioning. This dimension is consistent with the social amplification of risk framework, which argues that media can increase perceived threat by highlighting broader consequences, and with scholarship on mediatized disasters, where crises are often narrated as moments of structural fragility (Kasperson et al. 1988; Pantti et al. 2012; Altheide 2018). The headline *Valencia DANA leaves hospitals, roads, and transport completely paralyzed* refers a situation in which the disaster is framed as a systemic failure rather than a localized event. In this dimension, the emphasis shifts from the event itself to the failure of the systems expected to mitigate, contain or manage it.

Loss of Control/Inevitability refers to headlines that portray disasters as unstoppable, unavoidable, or beyond effective intervention. This dimension is closely related to research on fear communication and risk perception, which shows that hazards perceived as uncontrollable tend to generate stronger emotional responses and greater public concern than those viewed as manageable (Slovic 1987). Similarly, communication studies have demonstrated that messages emphasizing threat while minimizing efficacy or available responses can increase feelings of vulnerability and helplessness (Witte 1992). Since control perceptions play a central role in how individuals evaluate risk, headlines that emphasize the absence of effective intervention may amplify the perceived severity even when the factual circumstances remain unchanged. For example, the headline *Authorities unable to stop floodwaters as chaos spreads across Valencia* frames the event as exceeding institutional capacity and effective management. By removing agency from institutions, authorities, or affected populations, this dimension contributes to framing disasters as inevitable or uncontrollable outcomes rather than manageable crises.

Future Escalation refers to headlines that project the disaster toward further deterioration, suggesting that the situation is still worsening or may become more severe in the near future. This

dimension is consistent with the social amplification of risk framework, which argues that media communication can intensify public concern not only by emphasizing current impacts but also by highlighting the possibility of future consequences and expanding threats (Kasperson et al. 1988). It also aligns with research on news headlines as relevance optimizers, since references to anticipated developments increase the perceived urgency and significance of an event by directing attention toward what may happen next (Dor 2003; Molek-Kozakowska 2013). For example, the headline *The worst may still be ahead as new storms threaten Valencia* frames the disaster as an evolving threat whose consequences have not yet reached their peak. By emphasizing the prospect of further deterioration, this dimension amplifies uncertainty, urgency, and perceived risk beyond the current state.

Taken together, these dimensions capture recurrent rhetorical strategies through which media discourse may transform severe events into amplified threats. Rather than focusing solely on emotional language, the proposed framework identifies specific narrative structures that frame disasters as historically exceptional, systemically disruptive, uncontrollable, or likely to worsen. In this sense, rhetorical alarmism is conceptualized not as a property of isolated words or expressions, but as a multidimensional framing process that shapes how audiences interpret the magnitude, urgency, and evolution of a crisis. The four dimensions provide an operational basis for identifying and quantifying rhetorical alarmism in news headlines, enabling a systematic analysis for detecting its presence across media outlets.

4 Materials and Methods

4.1 Corpus Construction

This study analyzes a dataset of 1,347 headlines from different official news sources. The aim is to examine the presence of 'alarmist' framing in the coverage of the DANA event. The unit of analysis defined is the *journalistic headline* and in the case of social media, a sample for first *n-words* is taken from the post. The study period covers from 29th October to 30 November 2024, separating the data into two periods: the first from 29th October to 4th November (*Impact*), and the second from 5th November to 30th November (*Post*). This division allows for a comparative analysis of the framing used in the immediate aftermath of the DANA event and in the subsequent weeks.

To ensure an appropriate representation of Spanish media ecosystem, the corpus includes headlines from a variety of sources, which include national and local scopes. The data set includes headlines listed on the Table 1:

Data collection followed a reproducible procedure based on structured search queries. Whenever possible, the *Advanced Google Search Engine* was used, combining boolean operators and site domains. When this method was insufficient and provided poor results, internal search engines of media sites were used specifically in social media. A limitation was encountered in the case of *Instagram*, as the platform currently does not provide a reliable date-filtering mechanism comparable to search engines or other social media platforms such as *X* (Twitter). This constraint makes it difficult to reproduce the same sampling procedure, as result we decided to exclude this social media from the final corpus construction.

The general search pattern was as follows:

(dana OR "gota fría" OR lluvias OR torrenciales) AND Valencia AND 2024
site:<domain>

Explicit criteria were defined to ensure consistency across the headlines collected.

Inclusion Criteria

- Explicit reference to the phenomenon under study: DANA or heavy rainfall in Valencia

Media Type	Media Outlet
National Newspapers	<i>20 minutos, ABC, AS, EFE, El Mundo, El País EuropaPress, La Razón, La Vanguardia, Marca</i>
Local Newspapers	<i>Las Provincias, Levante EVM, El Mediterráneo, Torrent al día, Valencia Plaza</i>
National Radio	<i>La Cope, La Ser, Onda Cero</i>
Local TV	<i>7TeleValencia, À Punt, LaOchoTv</i>
National TV	<i>Antena 3, Telecinco, La Sexta, RTVE, Cuatro</i>
Social Media	<i>X, Facebook, Tiktok</i>

Table 1: Media outlets and social media included in the analysis

- Publication within the defined temporal window
- Headlines must originate from verifiable sources (URL or platform identifier)
- For social media: a minimum relevance threshold ($\geq 1,000$ interactions, such as likes or shares)

Exclusion Criteria

- Irrelevant or ambiguous content not directly related to the event
- Duplicated entries or republished content
- News outside the temporal scope

The sampling strategy consisted of selecting up to 50 relevant headlines per media outlet, ordered according to their position in search results and filtered based on the inclusion and exclusion criteria defined previously. This approach was adopted to ensure a comparable sample size across outlets, preventing larger media sources from being overrepresented relative to smaller or local outlets, and thereby enhancing the consistency and reproducibility of the analysis. In cases where a given outlet did not reach the target sample size, all available headlines meeting the selection criteria were retained.

A limitation of this approach is that the first set of sampled headlines does not necessarily correspond to equivalent temporal windows across media outlets. Due to differences in publication intensity over time, some outlets reached the target sample size more quickly than others. As a result, the collected samples may reflect not only outlet-specific framing strategies but also the natural temporal evolution of rhetorical alarmism.

Consequently, part of the observed differences in headline content may be driven by changes in the media discourse over time, rather than solely by differences between outlets. To address this issue, temporal variables such as time (in days) and coverage phase were incorporated into the statistical analysis. This allows for a more robust distinction between variations attributable to media-specific framing and those arising from the evolving phases of disaster reporting.

Following the data collection process, a structured database was constructed in which each observation corresponds to an individual headline. The final corpus comprises 1,347 headlines. For each observation, a set of variables was defined: the *media outlet* (source name), publication date, and the full headline text. Additionally, other categorical variables were incorporated, such as *media type* (e.g.,

press, radio, television, or social media) and the *temporal phase* of the event (impact or post-event stage). The dataset also includes an estimate of audience reach and the corresponding URL linking to the original content. The information related to audiences was collected from ...

4.2 Data Preprocessing

Text Normalization and Linguistic Processing

Prior to analysis, a text standardization process was applied to ensure consistency and reduce noise. The cleaning procedure included converting all text to lowercase, replacing punctuation with whitespace, collapsing multiple whitespace characters into a single space, and removing residual special characters. For social media content, additional preprocessing steps were implemented, including the removal of hashtags (#), mentions (@), emojis, and other non-linguistic symbols.

Once the text was normalized, all headlines were lemmatized to improve pattern detection. This process transforms each word (token) into its canonical base form, thereby minimizing the impact of inflectional variations such as verb conjugations, plural forms, and gender agreement. Lemmatization was performed using the UDPipe library (Straka and Straková 2017) with a Spanish pre-trained model (Taulé et al. 2008), which provides tokenization, part-of-speech tagging, and lemma extraction. Each headline was thus converted into a sequence of lemmas representing the original text. This lemmatized representation serves as the input for the subsequent regex-based detection process used to identify alarmist linguistic patterns.

Regex-based Algorithm for Detecting Rhetorical Alarmism

To classify news headlines according to the proposed framework, an automated algorithm was developed based on regular expression (*regex*) matching over lemmatized text. The construction of the regex patterns was guided by dictionaries associated with each of the four dimensions of *rhetorical alarmism* defined in this study. These dictionaries were manually elaborated and consist of keywords and linguistic expressions representative of each alarmism category. For each dimension, a set of regular expressions was generated to capture semantically and syntactically related patterns in the headlines. The algorithm evaluates each headline by matching its lemmatized tokens against the corresponding regex dictionaries. For every match, the associated alarmism category is recorded, the total number of detected alarmism categories is then computed for each headline.

To enable comparability across headlines of varying length, a normalized alarmism score is defined. Specifically, the number of detected alarmist patterns is divided by the total number of tokens in the headline, producing a dimensionless index of rhetorical alarmism (*RAI*). This normalization ensures that longer headlines are not systematically biased toward longer headlines, thereby allowing for fair comparisons across different media sources. Therefore, our index is defined in the Equation 4.2

$$RAI = \frac{N_m}{N_t}$$

Where N_m represent the number of matches of alarmist phrases and N_t total number of lemmatized words (tokens) of the headline.

For each dimension of rhetorical alarmism, we constructed a small and selective set of key patterns and phrases manually defined for DANA context. This restriction was intended to prevent inclusion of poorly related terms and to ensure that the detected patterns remained tightly aligned with each dimension. Accordingly, the dictionary were designed to capture salient and diagnostically relevant expressions, not broad lexical fields that might affect the measurement index. The Table 2 summarizes the *key* phrases and patterns defined for each dimension.

5 Conclusions

We conclude that....

Historical Exceptionality
<i>Key Patterns/Phrases:</i> Unprecedented; historic; record; never seen; unmatched; unparalleled; worst case/ever; extraordinary
Systemic Collapse
<i>Key Patterns/Phrases:</i> collapse; paralyz; breakdown; failure; malfunction; overload; gridlock; disintegrat; chaos; apocalyp
Loss of Control / Inevitability
<i>Key Patterns/Phrases:</i> uncontroll; unstoppable; uncontain; inevitable; unavoidable; overwhelm; out of control; helpless; beyond control
Future Escalation
<i>Key Patterns/Phrases:</i> worsen; escalat; intensif; increa; deteriorat; aggrav; upcoming; future threat; still to come; more to come

Table 2: Key phrases and word structures used to create the regex-based dictionary to identify rhetorical alarmism in headlines related to Valencia floods

External data

The experimetal dataset is available at....

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